

- 1) Задача (Установка): Установить Docker Desktop (для Win/Mac) или Docker Engine (для Linux).

```
root@Cetus:/home/ikomarov# docker --version
Docker version 28.5.1, build e180ab8
```

```
root@Cetus:/home/ikomarov# docker info
Client: Docker Engine - Community
Version: 28.5.1
Context: default
Debug Mode: false
Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version: v0.25.0
    Path: /usr/libexec/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version: v2.38.2
    Path: /usr/libexec/docker/cli-plugins/docker-compose

Server:
Containers: 0
  Running: 0
  Paused: 0
  Stopped: 0
Images: 8
Server Version: 28.5.1
Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d_type: true
  Using metacopy: false
  Native Overlay Diff: true
  userxattr: false
Logging Driver: json-file
Cgroup Driver: systemd
Cgroup Version: 2
Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
CDI spec directories:
  /etc/cdi
  /var/run/cdi
Swarm: inactive
Runtimes: runc io.containerd.runc.v2
Default Runtime: runc
Init Binary: docker-init
containerd version: b98a3aace656320842a23f4a392a33f46af97866
runc version: v1.3.0-0-g4ca628d1
init version: de40ad0
Security Options:
  apparmor
  seccomp
    Profile: builtin
  cgroupns
Kernel Version: 6.8.0-85-generic
Operating System: Ubuntu 24.04.2 LTS
OSType: linux
Architecture: x86_64
CPUs: 3
Total Memory: 3.824GiB
Name: Cetus
ID: 93896796-49e2-4830-b613-2a267ca015c1
Docker Root Dir: /var/lib/docker
Debug Mode: false
Experimental: false
Insecure Registries:
  ::1/128
  127.0.0.0/8
Live Restore Enabled: false
```

2) Задача ("Hello World"): Запустить первый контейнер: `docker run hello-world`.

```
root@Cetus:/home/ikomarov# docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
Digest: sha256:05813aedc15fb7b4d732e1be879d3252c1c9c25d885824f6295cab4538cb85cd
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

3) Задача (Интерактивный режим): Запустить интерактивный контейнер, чтобы "зайти" внутрь него: `docker run -it ubuntu bash`.

```
root@Cetus:/home/ikomarov# docker run -it ubuntu bash
root@fd9456c39c3:/# exit
exit
root@Cetus:/home/ikomarov#
```

4) Задача (Фоновый режим): Запустить веб-сервер nginx в фоновом (detached) режиме: `docker run -d nginx`.

```
root@Cetus:/home/ikomarov# docker run -d nginx
4bd2088197f6d97aeb0969e46249e69683bf3b7e6ee44b20e51bdcc1b084117d
```

5) Задача (Управление): Посмотреть запущенные контейнеры: `docker ps`.
Посмотреть все (включая остановленные): `docker ps -a`.

```
root@Cetus:/home/ikomarov# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
4bd2088197f6   nginx    "/docker-entrypoint..." 9 seconds ago  Up 8 seconds  80/tcp       recursing_raman

root@Cetus:/home/ikomarov# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
4bd2088197f6   nginx    "/docker-entrypoint..." 35 seconds ago  Up 34 seconds  80/tcp       recursing_raman
c4fabf459e34   ubuntu   "bash"                  4 minutes ago  Exited (0) 4 minutes ago  objective_ptolemy
42a2fad5f34f   hello-world "/hello"                4 minutes ago  Exited (0) 4 minutes ago  wonderful_diffie
```

- 6) Задача (Проброс портов): Остановить nginx (docker stop [id]). Запустить его заново с пробросом портов: docker run -d -p 8080:80 nginx.

```
root@Cetus:/home/ikomarov# docker stop 4bd2088197f6
4bd2088197f6
root@Cetus:/home/ikomarov# docker run -d -p 8080:80 nginx
cbb2fa384e40921084c251df542e8472f60747f29d85acd1e55d4ecc9d90385b
root@Cetus:/home/ikomarov# curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```


- 7) Задача (Логи): Посмотреть логи nginx: docker logs [id]. Посмотреть логи "в прямом эфире": docker logs -f [id].

```
root@Cetus:/home/ikomarov# docker logs cbb2fa384e40
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/01/26 21:49:08 [notice] 1#1: using the "epoll" event method
2026/01/26 21:49:08 [notice] 1#1: nginx/1.29.4
2026/01/26 21:49:08 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/26 21:49:08 [notice] 1#1: OS: Linux 6.8.0-85-generic
2026/01/26 21:49:08 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/01/26 21:49:08 [notice] 1#1: start worker processes
2026/01/26 21:49:08 [notice] 1#1: start worker process 29
2026/01/26 21:49:08 [notice] 1#1: start worker process 30
2026/01/26 21:49:08 [notice] 1#1: start worker process 31
172.17.0.1 - - [26/Jan/2026:21:49:27 +0000] "GET / HTTP/1.1" 200 615 "-" "curl/8.5.0" "-"
root@Cetus:/home/ikomarov# docker logs -f cbb2fa384e40
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/01/26 21:49:08 [notice] 1#1: using the "epoll" event method
2026/01/26 21:49:08 [notice] 1#1: nginx/1.29.4
2026/01/26 21:49:08 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/26 21:49:08 [notice] 1#1: OS: Linux 6.8.0-85-generic
2026/01/26 21:49:08 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/01/26 21:49:08 [notice] 1#1: start worker processes
2026/01/26 21:49:08 [notice] 1#1: start worker process 29
2026/01/26 21:49:08 [notice] 1#1: start worker process 30
2026/01/26 21:49:08 [notice] 1#1: start worker process 31
172.17.0.1 - - [26/Jan/2026:21:49:27 +0000] "GET / HTTP/1.1" 200 615 "-" "curl/8.5.0" "-"
```

- 8) Задача (Управление образами): Посмотреть скачанные образы: docker images.
Удалить образ: docker rmi hello-world.

```
root@Cetus:/home/ikomarov# docker images
REPOSITORY      TAG          IMAGE ID      CREATED        SIZE
ubuntu          latest       493218ed0f40  13 days ago   78.1MB
nginx           latest       4af177a024eb  13 days ago   161MB
my-web-app      latest       77cfb7d4fd4c  4 weeks ago   53.7MB
test_app        latest       7db53275b75e  5 weeks ago   153MB
nginx           alpine       04da2b0513cd  5 weeks ago   53.7MB
postgres        16-alpine    b97fe41a0657  5 weeks ago   276MB
hello-world     latest       1b44b5a3e06a  5 months ago  10.1kB
root@Cetus:/home/ikomarov# docker rmi hello-world
Untagged: hello-world:latest
Untagged: hello-world@sha256:05813aedc15fb7b4d732e1be879d3252c1c9c25d885824f6295cab4538cb85cd
Deleted: sha256:1b44b5a3e06a9aae883e7bf25e45c100be0bb81a0e01b32de604f3ac44711634
Deleted: sha256:53d204b3dc5ddbc129df4ce71996b8168711e211274c785de5e0d4eb68ec3851
root@Cetus:/home/ikomarov# docker images
REPOSITORY      TAG          IMAGE ID      CREATED        SIZE
ubuntu          latest       493218ed0f40  13 days ago   78.1MB
nginx           latest       4af177a024eb  13 days ago   161MB
my-web-app      latest       77cfb7d4fd4c  4 weeks ago   53.7MB
test_app        latest       7db53275b75e  5 weeks ago   153MB
nginx           alpine       04da2b0513cd  5 weeks ago   53.7MB
postgres        16-alpine    b97fe41a0657  5 weeks ago   276MB
```

- 9) Задача (Очистка): Остановить все контейнеры (docker stop \$(docker ps -q)).
Удалить все остановленные контейнеры (docker container prune -f).

```
root@Cetus:/home/ikomarov# docker stop $(docker ps -q)
cbb2fa384e40
root@Cetus:/home/ikomarov# docker container prune -f
Deleted Containers:
cbb2fa384e40921084c251df542e8472f60747f29d85acd1e55d4ecc9d90385b

Total reclaimed space: 1.093kB
root@Cetus:/home/ikomarov# docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS          NAMES
root@Cetus:/home/ikomarov#
```